ONCOLOGY · NEET-SS / FET / INI-SS

Small round cell tumours

1. Overview: the small round blue cell tumours

- On H&E these tumours look alike: sheets of **small, uniform cells with round nuclei and scant cytoplasm** ("small round blue cells"). Diagnosis depends on **immunohistochemistry, cytogenetics/molecular tests and age**, so biopsy tissue must be sent **fresh** (for FISH, RT-PCR, next-generation sequencing) as well as in formalin.
- **Members examined in orthopaedics:**
  - **Ewing sarcoma** (bone and soft tissue; includes the old "PNET" and **Askin tumour** of the chest wall)
  - **Primary lymphoma of bone**
  - **Multiple myeloma and solitary plasmacytoma**
  - **Metastatic neuroblastoma** (young children)
  - **Rhabdomyosarcoma** (alveolar type)
  - **Leukaemia** deposits
  - Others: **small cell osteosarcoma**, **mesenchymal chondrosarcoma**, **Langerhans cell histiocytosis** (covered with cystic lesions), osteomyelitis (inflammatory cells can mimic it).
- Mnemonic **LEMON** (Lymphoma, Ewing, Myeloma, Osteomyelitis/LCH, Neuroblastoma) is a common way to recall the differential for a permeative lesion.

Age as the first clue (classic exam rule)

Age	Most likely small round cell lesion of bone
Under 5 years	**Metastatic neuroblastoma**, leukaemia, LCH

Age	Most likely small round cell lesion of bone
5-30 years	**Ewing sarcoma** (peak 10-20 years)
Over 30-40 years	**Lymphoma** (adults of any age), **myeloma** (over 40-50), metastatic small cell lung carcinoma

Immunohistochemistry and genetics

Tumour	Markers	Genetic hallmark
Ewing sarcoma	**CD99 (MIC2) strong membranous**, **NKX2.2**, FLI1 nuclear; PAS positive (glycogen)	**t(11;22)(q24;q12) EWSR1-FLI1** (about 85%); t(21;22) EWSR1-ERG (about 10%)
Lymphoma (DLBCL)	**CD45 (leucocyte common antigen)**, **CD20** (B cell)	Variable
Myeloma / plasmacytoma	**CD138**, CD38, kappa or lambda light-chain restriction	t(4;14), t(14;16), del(17p) = high risk
Neuroblastoma	Synaptophysin, chromogranin, NSE; Homer Wright rosettes	**MYCN amplification** (poor prognosis)
Rhabdomyosarcoma	**Desmin, myogenin, MyoD1**	Alveolar: **t(2;13) PAX3-FOXO1**, t(1;13) PAX7-FOXO1
Mesenchymal chondrosarcoma	CD99 may be positive (pitfall), SOX9; islands of cartilage	**HEY1-NCOA2**
Small cell osteosarcoma	CD99 may be positive; **osteoid** production is the key	Not EWSR1-rearranged

Tumour	Markers	Genetic hallmark
Desmoplastic small round cell tumour	Desmin (dot-like) plus keratin	t(11;22)(p13;q12) EWSR1-WT1

- **WHO 2020** separates **"Ewing-like" round cell sarcomas** from true Ewing sarcoma: **CIC-DUX4 sarcoma**, **BCOR-CCNB3 sarcoma** and sarcomas with EWSR1 fused to non-ETS partners. They are mostly soft-tissue tumours, may be weakly CD99 positive, and CIC-rearranged sarcomas behave more aggressively.

> Exam pearl: CD99 is sensitive but not specific for Ewing sarcoma (also lymphoblastic lymphoma, synovial sarcoma, mesenchymal chondrosarcoma). The confirmatory test is demonstration of an EWSR1 rearrangement (FISH) or the EWSR1-FLI1 fusion transcript (RT-PCR), t(11;22) being the commonest.

2. Ewing sarcoma

Epidemiology

- **Second commonest primary malignant bone tumour in children and adolescents** (after osteosarcoma); overall about 3rd-4th commonest primary bone malignancy if myeloma is counted.
- **Age 5-25 years, peak in the second decade (10-15 years)**; rare under 5 and over 30.
- **Slight male predominance**. Much **commoner in people of European ancestry**, **rare in people of African ancestry** and less common in East Asians (germline polymorphisms near EGR2 that interact with the fusion protein).
- **Not hereditary**; no association with radiation or prior lesions (unlike osteosarcoma).
- **Cell of origin:** thought to be a mesenchymal stem cell; the EWSR1-FLI1 fusion acts as an aberrant transcription factor.

Sites

- **Long bones about 50%**: **femur is the commonest single bone**, then tibia, humerus, fibula. Classically **diaphyseal or metadiaphyseal** (contrast osteosarcoma: metaphyseal).
- **Flat bones about 40-50%**: **pelvis** (ilium most), ribs, scapula, sacrum, vertebrae. **Pelvis plus femur** account for about half of all cases.
- **Extraskeletal Ewing sarcoma** (about 15-20%): soft tissues of trunk and limbs; behaves similarly and is treated the same way.
- **Askin tumour:** Ewing sarcoma of the **thoracoabdominal wall** (chest wall, ribs).

Clinical features

- **Pain** (often intermittent at first, worse at night), **swelling**, palpable tender mass, warmth.
- **Systemic features in 10-20%: fever, weight loss, anaemia, raised ESR, raised WBC and raised LDH**, which makes it the **classic mimic of osteomyelitis**. Biopsy may yield "pus-like" liquefied necrotic tumour.
- Pathological fracture in about 10-15%.
- Rib or pelvic tumours present late with a large mass, pleural effusion or sciatica-like pain; spinal or sacral tumours with radicular symptoms or cord compression.

Radiology

- **Plain radiograph:**
  - **Permeative or moth-eaten lysis** (wide zone of transition) in the diaphysis.
  - **Lamellated "onion-skin" periosteal reaction** (repeated cycles of periosteal elevation).
  - **Codman triangle**; **"hair-on-end"** or sunburst spiculation in some.
  - **Large soft-tissue mass out of proportion to the bone destruction**.

  - **Saucerisation** (external erosion of the cortex by a subperiosteal mass) in periosteal-located tumours.
  - Sclerotic (reactive) changes in up to a third, especially pelvic tumours: do not assume osteosarcoma.
- **MRI of the whole bone** (with adjacent joints): extent of marrow and soft-tissue disease, **skip lesions**, neurovascular relations; plan biopsy and resection. The soft-tissue component often dwarfs the bone change.
- **Staging (systemic):**
  - **CT chest** (lungs are the commonest site of metastasis).
  - **Whole-body bone imaging: PET-CT** (now preferred; FDG-avid) or bone scintigraphy, whole-body MRI in some protocols.
  - **Bone marrow aspirate and trephine biopsy** (traditionally bilateral posterior iliac crests) because Ewing sarcoma spreads to marrow; many groups now omit it when PET-CT is negative (debated).
  - Bloods: FBC, ESR, **LDH** (prognostic), renal and liver function, fertility counselling before chemotherapy.
- **Biopsy:** core needle (image-guided) or open biopsy **planned with the definitive surgeon**, in line with the future incision, through a single compartment; soft-tissue component usually gives diagnostic tissue. Send **fresh tissue for molecular studies** and **culture** (to exclude infection).

Staging and prognosis

- Ewing sarcoma is by definition **high grade** (Enneking stage IIB if localised and extracompartmental, III if metastatic; AJCC staging also used).
- **About 20-25% have detectable metastases at diagnosis**: **lungs** (commonest), **bone and bone marrow**. Micrometastatic disease is assumed in all, hence chemotherapy for everyone.

- **Survival (approximate):** localised disease **about 70-75% at 5 years**; isolated lung metastases about 40%; bone or marrow metastases under 20-30%; relapse under 15%.

Prognostic factor	Unfavourable
Metastases at diagnosis	**Most important adverse factor**; bone/marrow worse than lung only
Site	**Pelvis, axial and proximal** sites worse than distal limb
Tumour size/volume	Over 8 cm or **over 200 mL**
Age	Older (over 15-18 years, adults) worse
Serum LDH	Raised
Histological response to induction chemotherapy	Poor response (**viable tumour 10% or more**, i.e. under 90% necrosis)
Fever, anaemia at presentation	Worse
Fusion type	Not prognostic with modern therapy (EWSR1-FLI1 vs ERG)

Management (multimodal; treat in a sarcoma centre)

1. **Neoadjuvant (induction) chemotherapy** for all patients after biopsy and staging.
2. **Local control** (surgery and/or radiotherapy) at about week 12-18, after reassessment imaging.
3. **Adjuvant (consolidation) chemotherapy**; total treatment about **9-12 months**.

Chemotherapy

- Backbone drugs: **vincristine, doxorubicin, cyclophosphamide, ifosfamide, etoposide** (actinomycin D in some regimens).
- **North American standard: VDC alternating with IE**. **COG AEWS0031 (Womer et al., 2012):** giving VDC/IE **every 2 weeks**

(interval-compressed) instead of every 3 weeks **improved event-free survival** without more toxicity.

- **Euro Ewing 2012 trial (Brennan et al., Lancet 2022): interval-compressed VDC/IE was superior to the European VIDE induction regimen** (better event-free and overall survival, less toxicity). VDC/IE is now the standard on both sides of the Atlantic.
- **High-dose busulfan-melphalan with autologous stem-cell rescue:** Euro-EWING 99/R2Loc (Whelan et al., 2018) showed better event-free survival than standard chemotherapy in **high-risk localised disease** (poor histological response or large tumour volume); no benefit in pulmonary metastases (R2Pulm).
- Late effects: cardiomyopathy (doxorubicin), infertility and haemorrhagic cystitis (alkylating agents), nephrotoxicity (ifosfamide), **secondary leukaemia/MDS** (etoposide, alkylators).

Local control: surgery versus radiotherapy

- **Ewing sarcoma is radiosensitive** (unlike osteosarcoma), so local control can be by surgery, radiotherapy or both. No randomised trial compares them; retrospective data favour **surgery when a wide margin can be achieved with acceptable function** (lower local recurrence), partly due to selection bias.

Option	Indications	Notes
Surgery alone (wide resection)	Resectable tumour, wide margins achievable, **good histological response**	**Preferred where feasible**; avoids radiation late effects. Reconstruction: endoprosthesis, allograft, vascularised fibula, rotationplasty, expendable bones (fibula, rib, clavicle) need no reconstruction

Option	Indications	Notes
Surgery plus postoperative radiotherapy	**Inadequate (marginal/intralesional) margins**, **poor histological response** (10% or more viable cells), some pelvic tumours	Doses around 45-55 Gy
Definitive radiotherapy alone	**Unresectable or functionally unacceptable resection**: spine, sacrum, many pelvic and skull-base tumours	About 55-56 Gy to the tumour; higher local recurrence than surgery in most series
Whole-lung irradiation	Lung metastases	Improves outcome in pulmonary metastatic disease

- **Radiotherapy complications: radiation-induced sarcoma** (commonly osteosarcoma, latency often over 5-10 years, dose-related), **physeal arrest and limb-length discrepancy**, **pathological fracture**, joint contracture, soft-tissue fibrosis, infertility (pelvic fields). This is why surgery is preferred in growing children when feasible.
- **Pathological fracture at presentation:** still treat with neoadjuvant chemotherapy first; splint or cast rather than internal fixation (fixation contaminates the whole bone); limb salvage remains possible.
- **Relapse:** early (under 2 years) relapse has a very poor prognosis; options include irinotecan-temozolomide, high-dose ifosfamide, high-dose chemotherapy with stem-cell rescue, and surgery or radiotherapy of oligometastases.

Differential diagnosis (distractors)

Condition	Distinguishing features
Acute or subacute osteomyelitis	Systemic illness more acute; MRI shows abscess, sequestrum; cultures positive; responds to antibiotics. **Always send biopsy for both culture and histology**
Langerhans cell histiocytosis	Age under 10, punched-out lesion, "vertebra plana", CD1a and langerin (CD207) positive
Osteosarcoma	Metaphyseal, **tumour osteoid** (cloud-like bone formation), raised ALP
Lymphoma of bone	Adults, CD45/CD20 positive, patient often well
Metastatic neuroblastoma	Under 5 years, raised urinary VMA/HVA, adrenal mass, MIBG avid
Stress fracture	Linear callus, history of activity, no soft-tissue mass

> Exam pearl: Ewing sarcoma = second decade, diaphysis of femur or pelvis, permeative lysis, onion-skin periosteal reaction, big soft-tissue mass, fever and raised ESR/LDH (mimics osteomyelitis), CD99 positive, t(11;22) EWSR1-FLI1. Worst prognostic factor = metastases at diagnosis. Standard chemotherapy = interval-compressed VDC/IE. Radiosensitive, but surgery is preferred for local control when a wide margin is achievable.

3. Primary lymphoma of bone

Definition and epidemiology

- **Primary bone lymphoma (PBL):** lymphoma arising in bone **without nodal or visceral disease** (classically no other site within 6 months of diagnosis). About **5% of extranodal lymphomas** and about **5-7% of primary malignant bone tumours**.
- **Secondary bone involvement** by systemic lymphoma (stage IV) is much commoner than primary bone lymphoma.

- **Histology: diffuse large B-cell lymphoma (DLBCL) in about 80% or more**; others follicular, anaplastic large cell, Hodgkin (rare in bone).
- **Adults, peak 40-60 years** (wide range, can occur in children), **male predominance**.
- **Sites: femur (commonest)**, pelvis, humerus, tibia, spine; metadiaphyseal.

Clinical and radiological features

- Bone pain, swelling, pathological fracture (about 20%), cord compression in spinal disease. **B symptoms (fever, night sweats, weight loss) uncommon**; the patient often looks **well relative to the size of the lesion**.
- **Radiographs: permeative or moth-eaten lysis** with a wide zone of transition, **little or no periosteal reaction**, sometimes mixed lytic-sclerotic; radiographs can look **nearly normal despite extensive marrow involvement** (cortex permeated by small channels rather than destroyed).
- **MRI:** extensive marrow replacement with a **large circumferential soft-tissue mass** around a relatively intact cortex.
- **Ivory vertebra:** Hodgkin lymphoma (differentials: osteoblastic prostate or breast metastasis, Paget disease).
- **Staging: PET-CT** (Lugano staging), bone marrow biopsy, CT chest/abdomen/pelvis, LDH.

Management

- **Biopsy** (core, with fresh tissue for flow cytometry); the diagnosis is made on CD20/CD45 positivity.
- **Chemotherapy: R-CHOP** (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone) for DLBCL, often followed by **consolidation radiotherapy** to the involved bone (benefit of adding radiotherapy is debated).

- **Surgery has no role in tumour removal.** Orthopaedic roles: **biopsy**, **fixation of pathological or impending fractures** (intramedullary nailing, cemented arthroplasty), decompression for cord compression when unstable.
- **Prognosis is good**: 5-year survival about 70-90% for localised DLBCL of bone, better than for most other primary bone malignancies.

> Exam pearl: An adult with a permeative lytic lesion, a large soft-tissue mass, minimal periosteal reaction and surprisingly few symptoms suggests primary lymphoma of bone. It is CD45/CD20 positive DLBCL, and it is treated with chemotherapy (R-CHOP) with or without radiotherapy, not resection.

4. Multiple myeloma and plasmacytoma

Epidemiology and pathology

- **Commonest primary malignant tumour arising in bone** (when included among bone tumours); bone involvement in about 80% at diagnosis.
- **Age over 40, median about 65-70 years**; **male predominance**; about **twice as common in people of African ancestry**.
- Clonal proliferation of **plasma cells** secreting a **monoclonal immunoglobulin (M-protein)**: **IgG commonest (about 50%)**, then **IgA (about 20%)**, light-chain-only (about 20%); IgD and non-secretory rare.
- Precursors: **MGUS** (progression about **1% per year**) and smouldering myeloma.
- **Bone destruction:** myeloma cells increase **RANKL** and reduce OPG, activating **osteoclasts**, and secrete **DKK1** (and sclerostin), which **suppresses osteoblasts**. Hence **purely lytic lesions, normal alkaline phosphatase, and a "cold" bone scan**.

- **Histology:** sheets of plasma cells (eccentric "clock-face" nucleus, perinuclear hof), **CD138 and CD38 positive**, kappa or lambda restricted. Amyloid may be present.

Clinical features (CRAB)

- **C**alcium raised (hypercalcaemia: confusion, constipation, polyuria).
- **R**enal impairment (light-chain cast nephropathy, hypercalcaemia, amyloid).
- **A**naemia (normocytic; marrow replacement): fatigue.
- **B**one disease: **back pain (commonest presenting symptom)**, vertebral collapse, pathological fractures, cord compression.
- Others: recurrent infections (immunoparesis), hyperviscosity, **amyloidosis** (carpal tunnel syndrome, macroglossia), peripheral neuropathy.

Diagnostic criteria (IMWG 2014)

- **Clonal bone marrow plasma cells 10% or more, or biopsy-proven bony or extramedullary plasmacytoma**, PLUS **one or more myeloma-defining events**:
  - **CRAB** features attributable to the plasma-cell disorder: calcium over 2.75 mmol/L (11 mg/dL), creatinine over 177 micromol/L (2 mg/dL) or clearance under 40 mL/min, haemoglobin under 10 g/dL (or 2 g/dL below normal), **one or more osteolytic lesions** (on CT, low-dose whole-body CT or PET-CT).
  - **SLiM biomarkers**: **Sixty** percent or more clonal plasma cells; **Light-chain** ratio (involved:uninvolved serum free light chain) 100 or more; **MRI** with more than one focal lesion of 5 mm or more.

Investigations

- **Bloods:** FBC, **ESR very high**, **rouleaux** on smear, calcium, creatinine, albumin, **beta-2 microglobulin**, LDH, **alkaline phosphatase normal**.

- **Serum protein electrophoresis with immunofixation** (M band), **serum free light-chain assay**, **urine for Bence Jones protein** (light chains; not detected by dipstick).
- **Bone marrow aspirate and trephine**, with cytogenetics/FISH.
- **Imaging:**
  - **Whole-body low-dose CT** is now preferred over the conventional skeletal survey (IMWG); **PET-CT** and **whole-body MRI** (spine and pelvis; detects marrow infiltration and SLiM focal lesions).
  - **Bone scan is unreliable (false negative)** because there is little osteoblastic activity: in suspected myeloma use radiographs/CT/MRI, not scintigraphy.
- **Radiographic signs: "punched-out" lytic lesions without a sclerotic rim**, **"raindrop" (pepper-pot) skull**, diffuse osteopenia, vertebral collapse (**pedicles usually spared early**, as the pedicle has less red marrow; contrast metastases, the "winking owl" sign), expansile lytic lesion in a rib or pelvis. **Sclerotic myeloma** occurs in **POEMS syndrome** (polyneuropathy, organomegaly, endocrinopathy, M-protein, skin changes).

Staging

System	Basis
ISS (International Staging System)	**Stage I:** beta-2 microglobulin under 3.5 mg/L and albumin 3.5 g/dL or more. **Stage III:** beta-2 microglobulin 5.5 mg/L or more. **Stage II:** neither I nor III
R-ISS (Revised ISS)	Adds **LDH** and **high-risk cytogenetics** (del(17p), t(4;14), t(14;16))
Durie-Salmon	Older system based on Hb, calcium, M-protein level and number of bone lesions

Management

- **Systemic therapy (haematology):** triplet or quadruplet regimens, e.g. **bortezomib (proteasome inhibitor), lenalidomide (immunomodulator), dexamethasone (VRd)**, often with the anti-CD38 antibody **daratumumab**; **autologous stem-cell transplantation** in fit patients; newer CAR-T and bispecific antibodies for relapse.
- **Bone-targeted therapy: zoledronic acid** (reduces skeletal events and has a modest survival benefit, MRC Myeloma IX) or **denosumab** (useful in renal impairment); calcium and vitamin D; watch for **osteonecrosis of the jaw**.
- **Avoid NSAIDs and nephrotoxic contrast**; hydrate.
- **Radiotherapy:** pain control, impending fracture, cord compression (myeloma is **very radiosensitive**; low doses, e.g. 8 Gy single fraction or 20 Gy in 5 fractions).
- **Orthopaedic and spinal surgery:**
  - **Biopsy any solitary lytic lesion** before fixation if the diagnosis is not established (a solitary lytic lesion may be myeloma, renal metastasis or primary sarcoma).
  - **Prophylactic fixation of impending fractures** and fixation of pathological fractures: **intramedullary nailing protecting the whole bone**, cemented arthroplasty for the femoral neck, cement augmentation. **Pathological fractures in myeloma have the highest healing rate** of all bone metastatic lesions (about two-thirds heal; Gainor and Buchert).
  - **Vertebral compression fractures: kyphoplasty or vertebroplasty** for painful fractures (**CAFE trial**, Berenson 2011: kyphoplasty improved pain and function in cancer-related fractures, mostly myeloma); brace.
  - **Cord compression:** dexamethasone and radiotherapy are first-line (radiosensitive); surgery for spinal instability, bony retropulsion or neurological deterioration despite radiotherapy.

Solitary plasmacytoma

- **Solitary plasmacytoma of bone:** a single lytic lesion of clonal plasma cells with **no CRAB features** and normal (or under 10% clonal) marrow.
- **Commonest in the spine (vertebral body)**, then pelvis, femur, ribs; younger than myeloma (median about 55).
- **Treatment: radiotherapy about 40-50 Gy** (local control over 85%); surgery for instability or fracture.
- **About 50-70% progress to multiple myeloma within 10 years**: follow with SPEP and free light chains.
- **Extramedullary plasmacytoma** (soft tissue, upper aerodynamic tract) has a lower rate of progression.

> Exam pearl: Myeloma lesions are purely lytic with normal ALP and a falsely negative bone scan, because osteoblasts are suppressed. Image with whole-body low-dose CT, PET-CT or MRI. A solitary lytic lesion in a patient over 40 must be biopsied before fixation.

5. Metastatic neuroblastoma

- **Commonest extracranial solid tumour of childhood**; **most present under 5 years (median about 18 months)**; arises from neural crest cells of the **adrenal medulla** or sympathetic chain.
- **Commonest cause of bone metastases in children under 5**; metastases to **skull and orbit (periorbital bruising, "raccoon eyes", proptosis)**, long-bone metaphyses (lytic, permeative, symmetric lesions), marrow ("Hutchinson" type spread to bone).
- **Clinical:** abdominal mass crossing the midline, limp, bone pain, irritability, fever (mimics osteomyelitis or leukaemia); opsoclonus-myoclonus syndrome; **Horner syndrome** (cervical/thoracic).
- **Investigations: raised urinary catecholamine metabolites (VMA and HVA)**; **MIBG scintigraphy** (iodine-123

metaiodobenzylguanidine) for primary and metastases; marrow biopsy.

- **Histology:** small round blue cells with **Homer Wright rosettes**, neuropil.
- **Prognosis:** better under 18 months; **MYCN amplification**, unfavourable histology (Shimada) and 11q loss are adverse. **Stage MS (formerly 4S)**: infants under 18 months with limited metastases (skin, liver, marrow) may regress spontaneously.
- **Spinal "dumbbell" tumour** through the neural foramen can cause cord compression and later post-laminectomy or post-radiation **kyphosis or scoliosis**.

6. Rhabdomyosarcoma and leukaemia (orthopaedic relevance)

Rhabdomyosarcoma

- **Commonest soft-tissue sarcoma of childhood**.

Type	Features
Embryonal (commonest)	Younger children (under 10), head and neck, genitourinary (**botryoid** variant: grape-like vaginal or bladder mass); better prognosis
Alveolar	**Adolescents, extremities** and trunk; **t(2;13) PAX3-FOXO1** or t(1;13) PAX7-FOXO1; worse prognosis; can metastasise to bone and marrow (mimics leukaemia) and lymph nodes
Pleomorphic	Adults; high grade

- **Treatment:** multi-agent chemotherapy (**VAC: vincristine, actinomycin D, cyclophosphamide**) plus local control by surgery and/or radiotherapy; **regional lymph-node evaluation** (sentinel node biopsy) for extremity tumours.

Acute leukaemia

- **ALL is the commonest childhood cancer**; up to about a third present with **bone pain, limp or arthralgia** and may be misdiagnosed as **juvenile idiopathic arthritis or osteomyelitis**.
- **Radiographic clues: metaphyseal lucent bands ("leukaemic lines")**, diffuse osteopenia, vertebral compression fractures, permeative lesions.
- **Red flags:** night pain, pain out of proportion to joint signs, cytopenias, blasts on smear, raised LDH and uric acid, hepatosplenomegaly, lymphadenopathy. **Do a peripheral smear before starting steroids for "JIA".**

7. Comparison of the main small round cell tumours of bone

Feature	Ewing sarcoma	Lymphoma of bone	Myeloma	Neuroblastoma metastasis
Age	5-25 (peak 10-15)	40-60 (any)	Over 40 (median 65-70)	Under 5
Site	Femur diaphysis, pelvis, ribs	Femur, pelvis, spine	Spine, skull, ribs, pelvis, proximal long bones	Skull, orbit, metaphyses
Radiograph	Permeative lysis, onion skin, big soft-tissue mass	Permeative, minimal periosteal reaction, big soft-tissue mass	Punched-out lysis, no sclerotic rim	Symmetric lytic metaphyseal lesions
Key test	CD99, EWSR1-FLI1	CD45, CD20	M-protein, CD138, marrow	Urine VMA/HVA, MIBG
Bone scan	Hot	Hot	**Often cold**	Hot (MIBG better)

Feature	Ewing sarcoma	Lymphoma of bone	Myeloma	Neuroblastoma metastasis
Mainstay of treatment	Chemo plus surgery and/or RT	Chemo (R-CHOP) plus or minus RT	Systemic therapy, bisphosphonate, RT	Chemo, surgery, RT, immunotherapy

Exam pearl: Age sorts the round cell tumours: neuroblastoma under 5, Ewing 5-25, lymphoma in middle age, myeloma after 40-50. Radiosensitive lesions in this group (Ewing, lymphoma, myeloma, plasmacytoma) explain why radiotherapy has a much larger role here than in osteosarcoma or chondrosarcoma.

Written by Claude for Ortho preSSaleel · updated 2026-11-15. Check doubtful points in a standard textbook.